



Massachusetts Institute of Technology

Fast methods for PDEs: Final Project

Spectral HPS scheme for Linear Elasticity

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Motivation

2D Linear Elasticity

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = \mathbf{0}, \quad \boldsymbol{\sigma} = 2\mu\boldsymbol{\epsilon} + \lambda(\text{tr } \boldsymbol{\epsilon})\mathbf{I}, \quad \boldsymbol{\epsilon} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$$

Navier-Lame formulation (component form for plane strain):

$$(2\mu + \lambda)\frac{\partial^2 u_x}{\partial x^2} + \mu\frac{\partial^2 u_x}{\partial y^2} + (\lambda + \mu)\frac{\partial^2 u_y}{\partial x\partial y} = -f_x$$

$$\mu\frac{\partial^2 u_y}{\partial x^2} + (2\mu + \lambda)\frac{\partial^2 u_y}{\partial y^2} + (\lambda + \mu)\frac{\partial^2 u_x}{\partial x\partial y} = -f_y$$

Two Benchmarks

- Smooth manufactured solution — spectral convergence?
- Mode I crack-tip singularity — algebraic convergence?

Tool

- Hierarchical Poincaré-Steklov spectral element solver
- Polynomial degree p , refinement depth L
- hp-convergence study: vary $p \in \{3, 4, 6, 8, 12, 16\}$ and $L \in \{2, 3, 4\}$

Acknowledgement

Library jaxhps by Melia et al., 2025
github.com/meliaoj/jaxhps

Method: Hierarchical Poincaré-Steklov (HPS)

Key Idea Instead of solving one big global system, solve many small spectral problems locally, then merge them hierarchically through boundary operators.

Two-Pass Solver

- 1 **Up-pass:** build Schur complements, leaf $\rightarrow$ root
- 2 **Down-pass:** recover solution, root $\rightarrow$ leaves

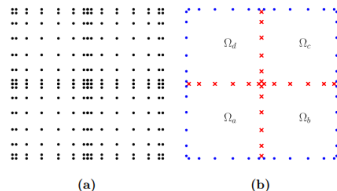
ADI (Alternating Direction Implicit)

Treat coupling term $\mathcal{L}_{xy} = (\lambda + \mu)\partial_{xy}$ as source and iterate:

$$\mathcal{L}_{xx} u_x^{(k+1)} = f_x - \mathcal{L}_{xy} u_y^{(k)}$$

$$\mathcal{L}_{yy} u_y^{(k+1)} = f_y - \mathcal{L}_{xy} u_x^{(k+1)}$$

- Each line is a scalar HPS solve with updated RHS.
- In some cases (rectangle), this converges in fixed number of iterations independent of discretization (Fortunato & Townsend, 2021).



Chebyshev grid on a leaf element (Melia et al., 2025)

Discretization

- Quadtree domain: L refinement levels
- Leaf K : $p \times p$ Chebyshev interior nodes

Problem 1: Smooth Manufactured Solution

Domain: $\Omega = [-1, 1]^2$, $\lambda = \mu = 1$

Exact solution

$$u_x = \sin(\pi x) \sin(\pi y)$$

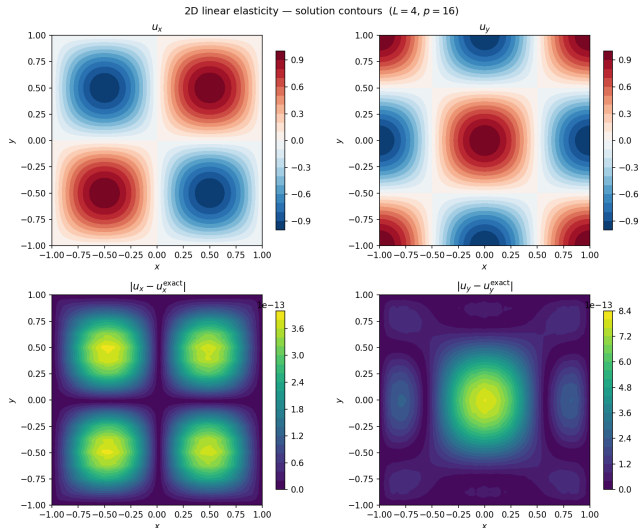
$$u_y = \cos(\pi x) \cos(\pi y)$$

Source term

$$f_x = 2\mu\pi^2 \sin(\pi x) \sin(\pi y)$$

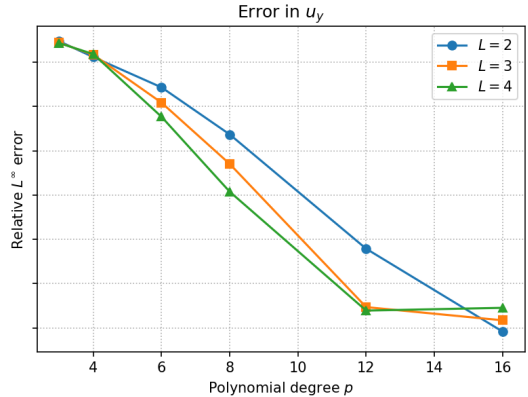
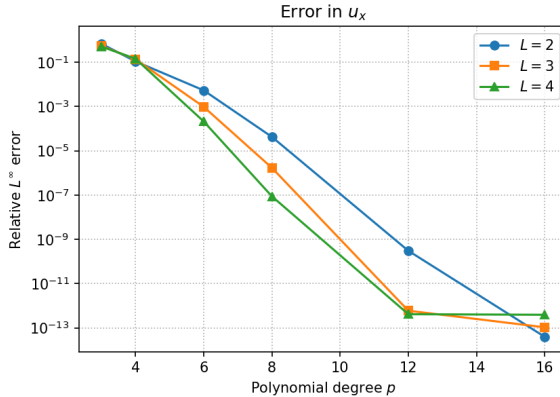
$$f_y = -(\lambda + 2\mu)\pi^2 \cos(\pi x) \cos(\pi y)$$

All derivatives analytic $\Rightarrow$ **exponential convergence**.



Problem 1: Convergence

2D linear elasticity — hp-convergence



Problem 2: Mode I Crack-Tip Singularity

Williams Expansion (Mode I, plane strain)

$$u_x = \frac{K_I}{2\mu} \sqrt{\frac{r}{2\pi}} \cos \frac{\theta}{2} (\kappa - \cos \theta)$$

$$u_y = \frac{K_I}{2\mu} \sqrt{\frac{r}{2\pi}} \sin \frac{\theta}{2} (\kappa - \cos \theta)$$

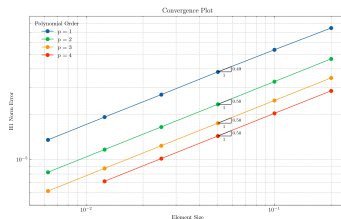
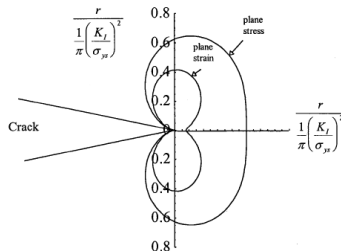
$$\kappa = 3 - 4\nu$$

Stress singularity

$$\sigma(r, \theta) = \frac{K_I}{\sqrt{2\pi r}} \mathbf{f}(\theta) + O(r^{1/2})$$

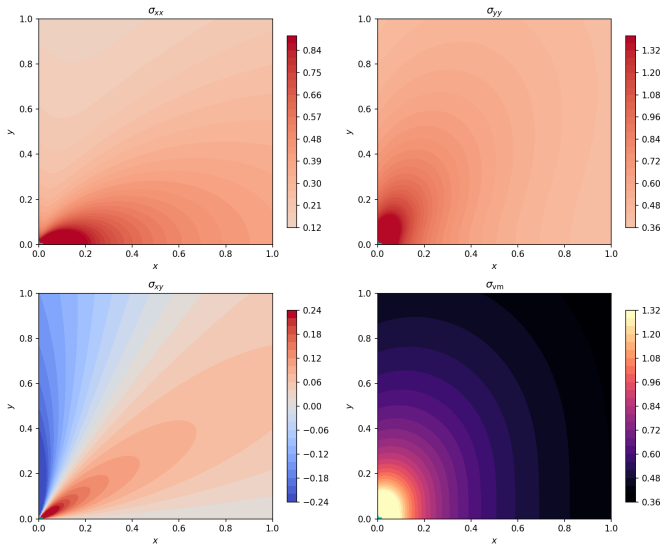
$$\nabla u \sim r^{-1/2} \text{ as } r \rightarrow 0^+$$

Dirichlet BCs from exact solution.

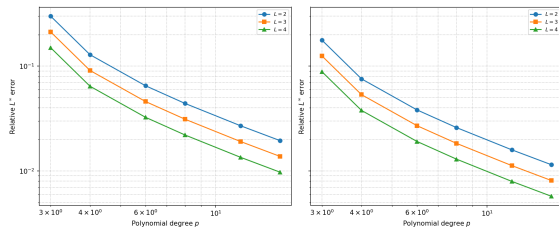


FEM convergence on crack-tip problem

Problem 2: Stress Fields

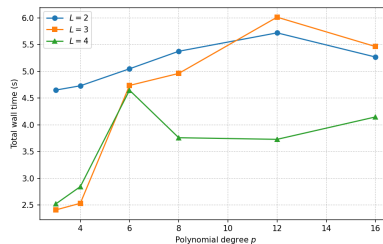
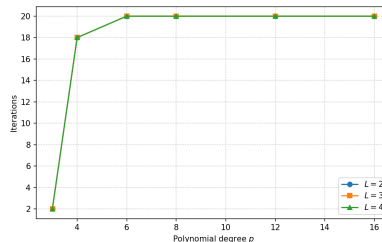


Problem 2: Convergence, Iterations & Runtime



Relative L^∞ error vs. polynomial degree

- Algebraic convergence (singularity-limited)
- Increasing L (refinement) helps
- Iteration count appears independent of p



Conclusions

Main Results

- Smooth elasticity case: spectral convergence
- Crack-tip case: algebraic convergence due to $\frac{1}{\sqrt{r}}$ singularity
- ADI coupling converges seems to be independent of refinement and polynomial degree
- HPS remains efficient as p and L increase

Next steps:

- Adaptive refinement near the crack tip singularity
- Investigate incompressible limit ($\nu \rightarrow 0.5$, $\lambda \rightarrow \infty$)